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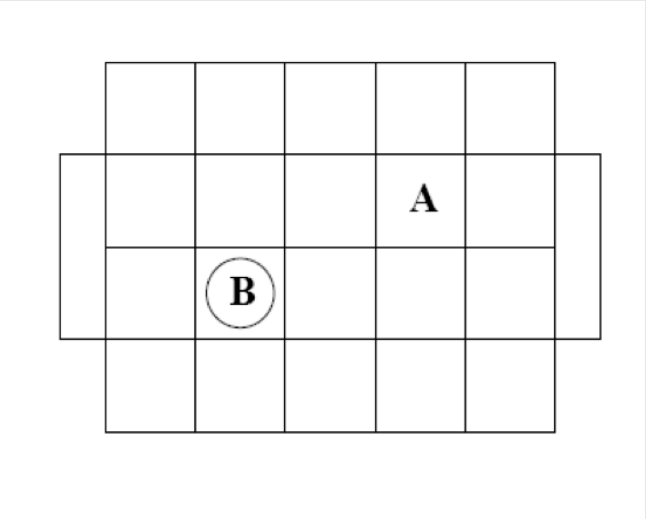
MONASH University
Information Technology

FIT5226
Multi-agent System & Collective Behaviour

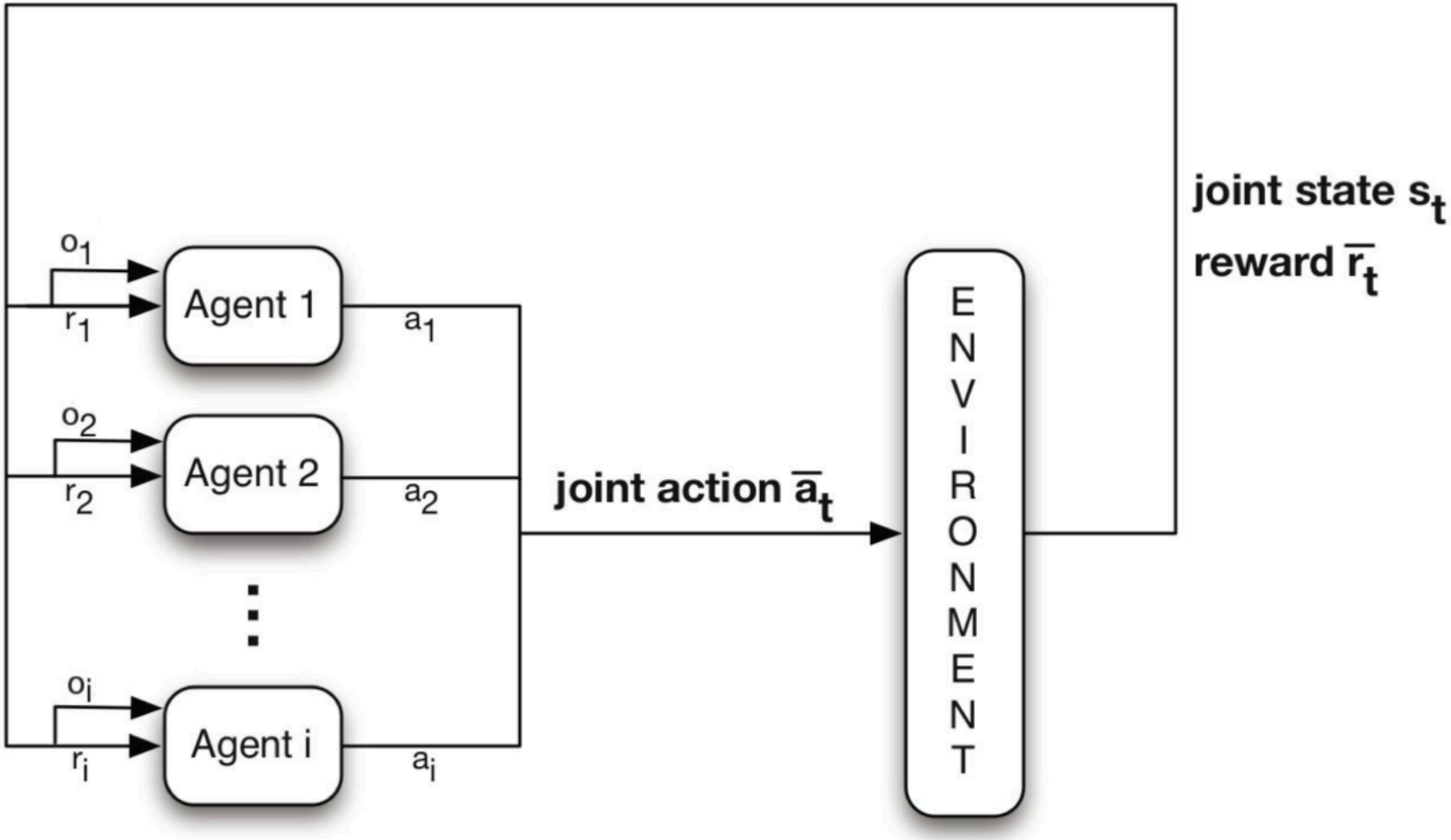
Wk 7: MiniMax Q-Learning

Bernd Meyer, September 2024

Robo Soccer



Multi-Agent-Environment Interaction



Source: Nowe, Vrancx & De Hauwere 2012

Markov Games

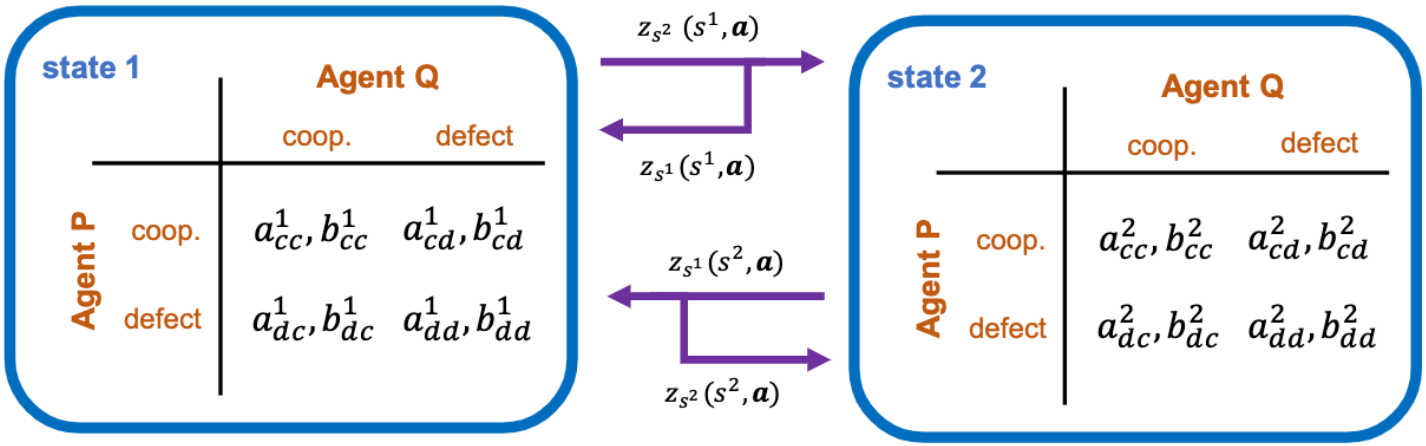
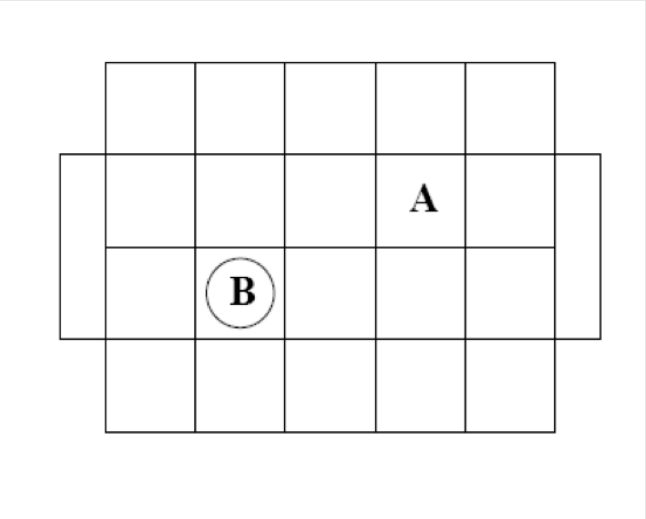


Figure 1: The framework of a two-state two-player two-action stochastic game. There are two players, P and Q. In every round, they can be in one state, s^1 or s^2. In each state, each player has a corresponding strategy and payoff matrix. The state in the next round depends on the current state, on the players' joint action and on chance, $z_{s'}(s, \boldsymbol{a})$.

Diagram from C. Deng et al: *Modeling Replicator Dynamics in Stochastic Games Using Markov Chain Method.* AAMAS 2021

Example of a Markov Game



Markov Game: Definition

Definition 6.2.1 (Stochastic game) *A* stochastic game *(also known as a* Markov game*) is a tuple* (Q, N, A, P, r)*, where:*

- Q *is a finite set of games;*
- N *is a finite set of* n *players;*
- $A = A_1 \times \cdots \times A_n$*, where* A_i *is a finite set of actions available to player* i*;*
- $P : Q \times A \times Q \mapsto [0, 1]$ *is the transition probability function;* $P(q, a, \hat{q})$ *is the probability of transitioning from state* q *to state* $\hat{q}$ *after action profile* a*; and*
- $R = r_1, \ldots, r_n$*, where* $r_i : Q \times A \mapsto \mathbb{R}$ *is a real-valued payoff function for player* i*.*

Nash in Markov Games

Theorem:

Every two-player, constant-sum, average reward, irreducible stochastic game has a Nash equilibrium

"irreducible" means here that the Markov chain of games must be irreducible, in other words, every game must always remain reachable, regardless of which strategies are played.

Standard Q-Learning

Recall standard Q-Learning (here in the formulation from SLB).

This can be adapted to 2-player zero-sum games relatively immediately.

Definition 7.4.1 (Q-learning) *Q-learning is the following procedure:*

Initialize the Q-function and V values (arbitrarily, for example)
repeat *until convergence*
- Observe the current state s_t.
- Select action a_t and and take it.
- Observe the reward $r(s_t, a_t)$
- Perform the following updates (and do not update any other Q-values):
- $Q_{t+1}(s_t, a_t) \leftarrow (1-\alpha)Q_t(s_t, a_t) + \alpha_t(r(s_t, a_t) + \beta V_t(s_{t+1}))$
- $V_{t+1}(s) \leftarrow \max_a Q_t(s, a)$

Q-value derivation for known *joint* action

$$Q_{i,t+1}(s_t, a_t, o_t) = (1 - \alpha_t)Q_{i,t}(s_t, a_t, o_t) + \alpha_t(r_i(s_t, a_t, o_t) + \beta V_t(s_{t+1}))$$

a_t is the player's own action, o_t that of the opponent.

Nothing has changed except for switching from a single agent's action to the joint action. This means that the player needs to know the other player's action to perform MiniMax-Q.

State value update according to MiniMax

The core of the method is to calculate the value according to the standard MiniMax method for 2-player zero-sum games.

$$V_t(s) = \max_{\Pi_i} \min_{o} Q_{i,t}(s, \Pi_i(s), o).$$

In practice, this requires the use of a linear solver within the Q-iteration.

The MiniMax-Q algorithm

// Initialize:

forall $s \in S, a \in A$, *and* $o \in O$ **do**

$\lfloor\ Q(s,a,o) \leftarrow 1$

forall s *in* S **do**

$\lfloor\ V(s) \leftarrow 1$

forall $s \in S$ *and* $a \in A$ **do**

$\lfloor\ \Pi(s,a) \leftarrow 1/|A|$

$\alpha \leftarrow 1.0$

// Take an action:

when in state s, with probability $explor$ choose an action uniformly at random, and with probability $(1 - explor)$ choose action a with probability $\Pi(s,a)$

// Learn:

after receiving reward rew for moving from state s to s' via action a and opponent's action o

$Q(s,a,o) \leftarrow (1-\alpha) * Q(s,a,o) + \alpha * (rew + \gamma * V(s'))$

$\Pi(s,\cdot) \leftarrow \arg\max_{\Pi'(s,\cdot)}(\min_{o'} \sum_{a'}(\Pi(s,a') * Q(s,a',o')))$

// The above can be done, for example, by linear programming

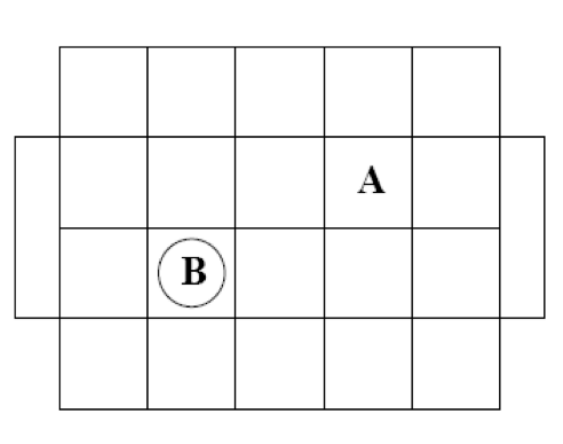
$V(s) \leftarrow \min_{o'}(\sum_{a'}(\Pi(s,a') * Q(s,a',o')))$

Update α

Figure 7.8: The minimax-Q algorithm.

Example: Grid Soccer Defense

- 2-player soccer
- 4x5 grid
- 5 actions (N, S, E, W, stop)



- Four policies trained:
  - Minimax-Q trained against random opponent
  - Minimax-Q trained against Minimax-Q opponent
  - Q trained against random opponent
  - Q trained against Q opponent

	MR		MM		QR		QQ	
	% won	games	% won	games	% won	games	% won	games
vs. random	99.3	6500	99.3	7200	99.4	11300	99.5	8600
vs. hand-built	48.1	4300	53.7	5300	26.1	14300	76.3	3300
vs. MR-challenger	35.0	4300						
vs. MM-challenger			37.5	4400				
vs. QR-challenger					0.0	5500		
vs. QQ-challenger							0.0	1200

Convergence and Limitations

Theorem 7.4.3 *Under the same conditions that assure convergence of Q-learning to the optimal policy in MDPs, in zero-sum games Minimax-Q converges to the value of the game in self play.*

- **MiniMax-Q** converges but only in 2-player zero-sum Games
- **Friend-or-Foe-Q** extends this to general sum *n*-player games, but the other players must be known to the agent as either "friend" (playing for coordinated equilibrium) or "foe" (playing for adversarial equilibrium), so in a sense this is a 2-player game again, just in teams.
- More general algorithms exist but they are only guaranteed to converge only under very strict conditions
- No known solution for general-sum, *n*-player games